BOILEAU GUILLAUME

Senior Researcher / Quantitative Climate Risk | Python, modelling, uncertainty, physical risk & applied research

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EXPERIENCE

Software Engineer – Data workflows, operational risk & technical documentation

VEV Platform Services France

2025 – 2026 · Valbonne, France

Context: Industrial software platform connected to electric vehicle charging infrastructure, with operational data flows, field constraints, incident analysis and digital service reliability.
Objective: Contribute to the reliability of operational digital services by analysing data flows, incidents, application behaviours and the interface between software, infrastructure and field operations.

- **Operational data analysis:** Investigation of FTP data exchanges, interrupted flows, data inconsistencies and unexpected application behaviour.
- **Risk and incident analysis:** Diagnosis of operational incidents, root-cause analysis, formalisation of findings and support to corrective actions.
- **Infrastructure-related systems:** Work at the interface between software, field infrastructure, operational data and service constraints.
- **Technical documentation:** Production of clear technical notes, anomaly analyses and structured information for engineering teams.
- **Software contribution:** Development contributions in TypeScript, Angular and Node.js within an operational platform.
- **Collaborative tools:** Use of Linux, Git, Zenhub and technical documentation workflows.

Senior R&D Engineer – Quantitative modelling, uncertainty & reproducible pipelines

CNES

2023 – 2025 · Nice, France

Context: Applied R&D and software development for the LISA ESA/NASA space mission ground segment, involving mission data chains L0–L3, simulation, model integration, uncertainty, validation and reproducible scientific workflows.
Objective: Design robust Python workflows to integrate, compare, validate and document complex models and heterogeneous data, with emphasis on uncertainty, traceability, reproducibility and methodological rigour.

- **Quantitative research workflows:** Development of CATpyLISA, a Python platform for model integration, comparison, validation and technical review. GitLab: CATpyLISA
- **Methodology development:** Structuring of assumptions, input datasets, configurations, intermediate results, model outputs and validation metrics.
- **Uncertainty and sensitivity:** Analysis of weak signals, noise contributions, parameter effects, limitations, model behaviour and robustness.
- **Model validation:** Definition of acceptance criteria, comparison rules, consistency checks, validation procedures and gap analyses.
- **Reproducible pipelines:** Design of documented workflows enabling traceable analyses, reviewability and collaborative use.
- **Applied research outputs:** Preparation of technical notes, review material, methodological explanations and decision-support summaries.
- **International collaboration:** Interface with scientific, software and institutional contributors in the LISA international context.

Data Scientist – Physical risk proxies, sensor data & statistical modelling

Universiteit Antwerpen, Belgium

2021 – 2023 · Antwerp, Belgium

Context: Applied data science for precision instruments and next-generation detectors, combining sensor data, environmental noise, statistical modelling, machine learning and performance validation.
Objective: Develop robust workflows to analyse sensor datasets, characterise environmental contributions, model uncertainties, compare configurations and produce decision-oriented recommendations.

- **Sensor and environmental data:** Analysis of environmental measurements, spatial correlations, propagation effects and their impact on system performance.
- **Physical risk reasoning:** Assessment of how external physical phenomena propagate into system-level performance degradation and uncertainty.
- **Statistical modelling:** Development of MCMC pipelines for different configurations and noise assumptions.
- **Machine learning:** Development of approaches for non-stationary noise reduction using witness channels, machine learning and deep learning.
- **Performance metrics:** Figures of merit, sensitivity analyses, consistency checks and comparison of configurations.
- **Research communication:** Production of traceable results, technical syntheses and recommendations for international stakeholders.

PROFILE

PhD-level quantitative researcher with strong experience in applied modelling, uncertainty analysis, Python pipelines, reproducible research, physical-risk reasoning, model validation and scientific publication.
Target position: contribute to EDHEC Climate Institute's real estate climate risk methodology by bringing rigorous quantitative modelling, transparent assumptions, documented pipelines, uncertainty analysis and a strong capacity to bridge research outputs with operational decision-making.

- Strong research foundation: PhD in physics, peer-reviewed scientific output, applied R&D and international collaboration.
- Robust quantitative toolkit: Python, statistical modelling, simulation, model validation, uncertainty, sensitivity analysis and reproducible pipelines.
- Relevant physical-risk mindset: experience modelling how external physical phenomena, environmental noise and system vulnerabilities affect performance.
- Strong writing ability for scientific, technical and practitioner-oriented audiences.
- Targeted ramp-up on real estate finance, climate finance, CRREM, EPC datasets, LTV, PD/LGD, SFDR, CSRD and EBA guidelines.

LANGUAGES

English ●●●●○

Spanish ●●○○○

EDUCATION

PhD in Physics

Université Côte d'Azur

2018 – 2021 · Nice, France

Selective Graduate Program in Fundamental Physics (Magistère)

Université Paris-Saclay

2016 – 2018 · Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

2017 – 2018 · Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

2015 – 2016 · Orsay, France

PROFESSIONAL TRAINING

Machine Learning in Python with scikit-learn

FUN MOOC – Inria

2026 · France Université Numérique

Predictive modelling, supervised learning, model evaluation, cross-validation, metrics, pipelines and methodological/software fundamentals of machine learning in Python.

Junior R&D Engineer – Scientific modelling, simulation & publications

ARTEMIS, Observatoire de la Côte d'Azur

2018 – 2021 | Nice, France

Context: Applied research on weak-signal analysis, numerical simulation, signal processing and preparation of space mission science cases.
Objective: Develop modelling and simulation methods to produce robust, documented and publishable scientific results.

- **Scientific modelling:** Development of numerical simulation workflows for complex signal environments and large datasets.
- **Signal processing:** Methods to analyse and separate weak, overlapping signals.
- **Validation:** Consistency checks, result comparison, sensitivity studies and methodological limitation analysis.
- **Academic writing:** Contribution to scientific publications, reports, methodological explanations and presentations.

Reproducible Research: Methodological Principles for Transparent Science

FUN MOOC – Inria

2025 | France Université Numérique

Reproducible research, GitLab, notebooks/Jupyter, data management, computational documentation, software environments and reproducibility methods.

Continuous Professional Training – Data, AI, Software and Systems

OpenClassrooms / Coursera / FUN MOOC

2020 – 2026

Python, SQL, Machine Learning, AI, Linux, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, C/C++, data analysis and software engineering fundamentals.

PROJECTS

CATpyLISA – Reproducible modelling and validation platform

Python / Scientific workflows / Model comparison

2023 – 2025

Python platform for integrating models, structuring data, comparing results, validating outputs and supporting collaborative technical reviews in a complex scientific environment.
Role: Python development, model validation, data structuring, reproducible workflows, sensitivity checks, gap analysis, documentation and coordination with multiple contributors.
Transfer to climate risk: methodology development, transparent assumptions, reproducible pipelines, model validation, uncertainty analysis and practitioner-oriented documentation.

Environmental data, risk propagation and performance modelling

Sensor data / Physical risk / Statistical modelling

2021 – 2023

Development of numerical and statistical workflows to analyse environmental sensor data, characterise physical disturbances, compare configurations and assess performance impacts.
Role: sensor data analysis, statistical modelling, MCMC pipelines, non-stationary noise analysis, performance metrics, validation and reporting.
Transfer to real estate climate risk: reasoning from physical shocks to system-level vulnerability, uncertainty, scenario comparison and decision-support metrics.

Scientific publications and applied research outputs

Peer-reviewed research / Modelling / Simulation

2018 – 2025

Scientific work involving modelling, simulation, uncertainty, validation, weak-signal analysis and peer-reviewed publication in international research environments.
Role: modelling, numerical analysis, scientific writing, result validation, interpretation, communication and collaborative review.
Transfer to EDHEC Climate Institute: academic rigour, ability to publish, build intellectual presence and translate quantitative frameworks into usable outputs.

CORE SKILLS

Quantitative research & modelling

Quantitative modelling · Applied research · Methodology development · Uncertainty analysis · Sensitivity analysis · Scenario analysis · Statistical modelling · MCMC · Model validation · Assumption testing · Boundary conditions · Scientific publication

Climate risk & real estate finance

Climate risk – ramp-up · Physical risk · Transition risk – ramp-up · Real estate finance – ramp-up · DCF – ramp-up · Income capitalisation – ramp-up · LTV – ramp-up · PD/LGD – ramp-up · Stranded assets – awareness · Insurance availability – awareness · Retrofit CapEx – awareness · Regulatory costs – awareness

Built environment data & frameworks

EPC datasets – ramp-up · Energy labels – ramp-up · Building stock models – ramp-up · Building archetypes – ramp-up · CRREM – ramp-up · LCA frameworks – ramp-up · Data coverage · Data quality · Calibration strategies · Substitution strategies · Portfolio datasets – ramp-up

Python, data & reproducibility

Python · pandas · NumPy · SciPy · Matplotlib · Jupyter · SQL · Git / GitLab · Linux · Reproducible pipelines · Documentation · Data analysis · Data validation

Regulation, finance stakeholders & communication

SFDR – awareness · CSRD – awareness · EBA guidelines – awareness · Financial institutions – ramp-up · Regulators – awareness · Client-facing notes · Academic writing · Policy forums – ramp-up · Industry workshops · Cross-disciplinary collaboration · English

Systems, risk & decision support

Risk transmission · Vulnerability modelling · Portfolio-level reasoning · Performance metrics · Decision-support tools · Complex systems · Physical processes · Validation · Robustness · Technical synthesis · Stakeholder communication